

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8							
A																					
B																					
C																					
D																					
E																					
F																					
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM SBC-W OUTGOING H02																					
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV							
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		COVER SHEET		Scale		NTS		= H02	
00		Issued for Approval		17.01.19		YR		Checked		AW				MVD-32 20kV Tank Farm WICA		Customer		PT Multimas Nabati Asahan		& 306	
														SCHEMATIC DIAGRAM		Drawing Nbr.		18.042-MNA-306.11		Sh. 1	
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318042				SBC-W OUTGOING		Customer Doc. Nbr.		Cont. 27	
1		2		3		4		5		6		7		8							

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

SHEET INDEX

PAGE	PAGE NAME	REVISION		
==MVD-31=H02,H03,H04&306/1	COVER SHEET	00		
==MVD-31=H02,H03,H04&306/2	Index of Sheet	00		
==MVD-31=H02,H03,H04&306/3	REFERENCE DESIGNATIONS	00		
==MVD-31=H02,H03,H04&306/4	REFERENCE DESIGNATIONS	00		
==MVD-31=H02,H03,H04&306/5	REFERENCE DESIGNATIONS	00		
==MVD-31=H02,H03,H04&306/6	REFERENCE DESIGNATIONS	00		
==MVD-31=H02,H03,H04&306/7	DISTRIBUTION OF AUXILIARY CIRCUITS	00		
==MVD-31=H02,H03,H04&306/8	AUXILIARY SUPPLY CIRCUIT	00		
==MVD-31=H02,H03,H04&306/9	CIRCUIT BREAKER CONTROL	00		
==MVD-31=H02,H03,H04&306/10	AUXILIARY MOTOR & RELAY	00		
==MVD-31=H02,H03,H04&306/11	DIGITAL INPUT/OUPUT RELAY	00		
==MVD-31=H02,H03,H04&306/12	SPARE CONTACT	00		
==MVD-31=H02,H03,H04&306/13	CURRENT TRANSFOMER	00		
==MVD-31=H02,H03,H04&306/14	VOLTAGE TRANSFORMER	00		
==MVD-31=H02,H03,H04&306/15	LV DOOR LAYOUT	00		
==MVD-31=H02,H03,H04&306/16	LV BOX LAYOUT	00		
==MVD-31=H02,H03,H04&306/17	PART LIST	00		
==MVD-31=H02,H03,H04&306/18	Terminal diagram ==MVD-31=H02,H03,H04+-XDA			
==MVD-31=H02,H03,H04&306/19	Terminal diagram ==MVD-31=H02,H03,H04+-XDB1			
==MVD-31=H02,H03,H04&306/20	Terminal diagram ==MVD-31=H02,H03,H04+-XDC1			
==MVD-31=H02,H03,H04&306/21	Terminal diagram ==MVD-31=H02,H03,H04+-XDC2			
==MVD-31=H02,H03,H04&306/22	Terminal diagram ==MVD-31=H02,H03,H04+-XDI			
==MVD-31=H02,H03,H04&306/23	Terminal diagram ==MVD-31=H02,H03,H04+-XDI2			
==MVD-31=H02,H03,H04&306/24	Terminal diagram ==MVD-31=H02,H03,H04+-XDI3			
==MVD-31=H02,H03,H04&306/25	Terminal diagram ==MVD-31=H02,H03,H04+-XDI6			
==MVD-31=H02,H03,H04&306/26	Terminal diagram ==MVD-31=H02,H03,H04+-XDV			
==MVD-31=H02,H03,H04&306/27	Terminal diagram ==MVD-31=H02,H03,H04+-XDX			

			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	Index of Sheet MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02	
			Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306	
00	Issued for Approval	17.01.19	YR	Checked		AW	Drawing Nbr.		18.042-MNA-306.11	Sh.	2	
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		Customer Doc. Nbr.		Cont.	27
						PT. ABB Sakti Industri						

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

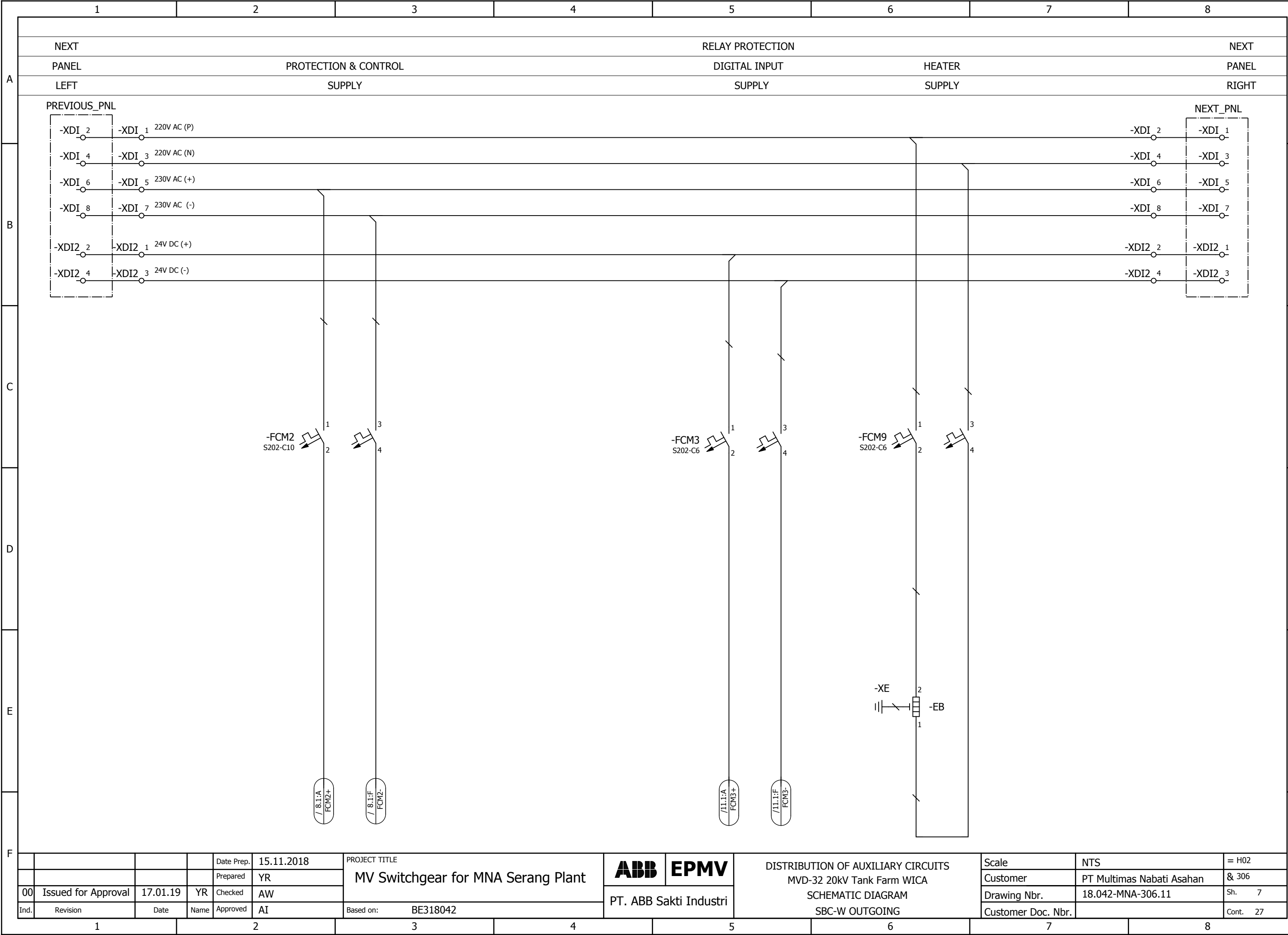
1		2		3		4		5		6		7		8		
A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
B	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							B
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
C	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							C
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
D	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							D
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
E	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							E
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
F	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							F
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
G	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							G
	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
H	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							H
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
I	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							I
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
J	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							J
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
K	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							K
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
L	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							L
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
M								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							M
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
1		2		3		4		5		6		7		8		
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-32 20kV Tank Farm WICA		Scale	NTS		= H02	
				Prepared	YR							Customer	PT Multimas Nabati Asahan		& 306	
00	Issued for Approval	17.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBC-W OUTGOING		Drawing Nbr.	18.042-MNA-306.11		Sh. 4	
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042					Customer Doc. Nbr.			Cont. 27	

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

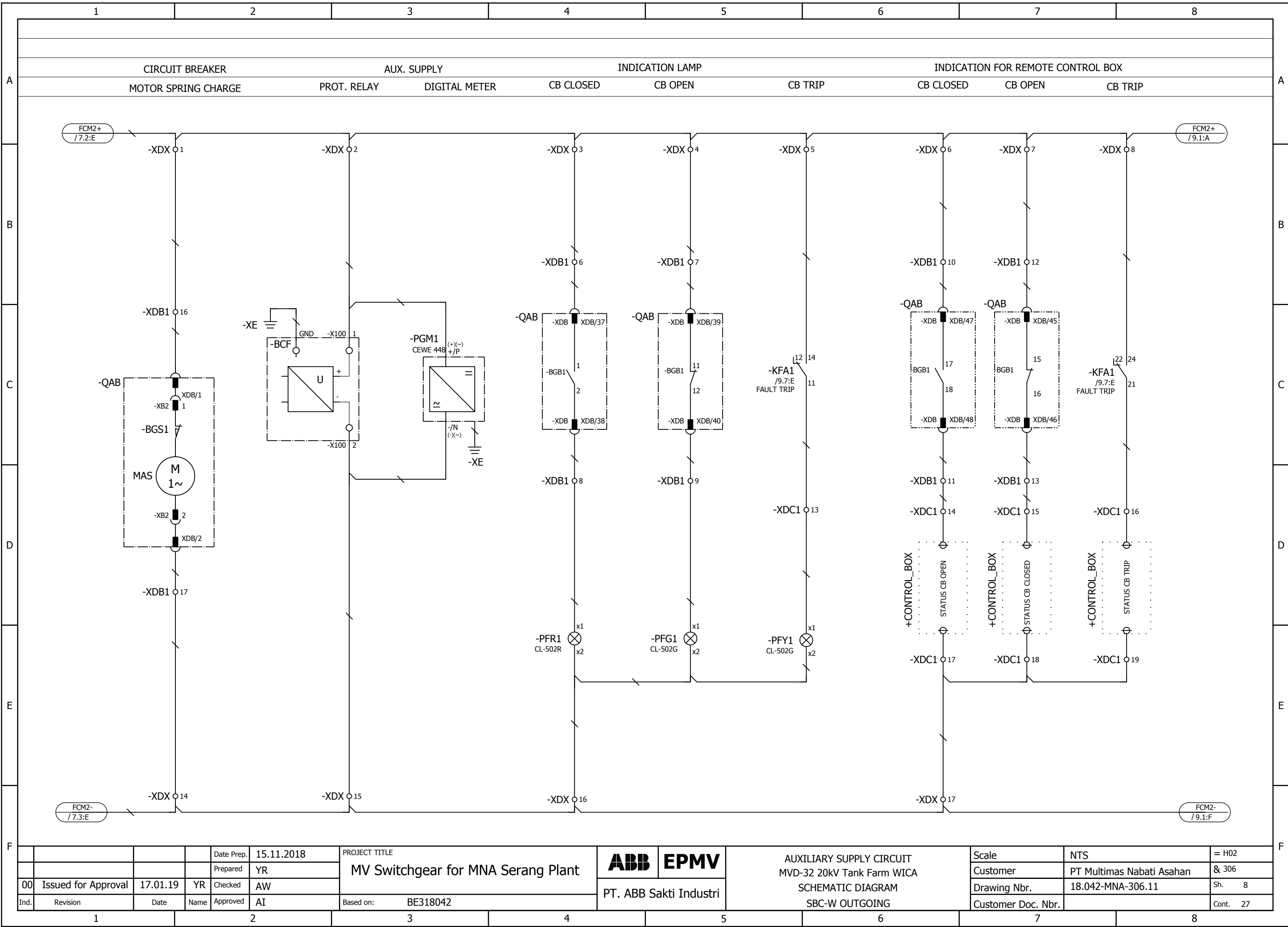
1		2		3		4		5		6		7		8				
A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS							A		
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS									
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS									
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS									
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK									
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING									
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH									
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES									
	B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR							B	
		-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES								
-PFB		BLUE SIGNAL LAMPS						-RAI	INDUCTORS									
-PFF		FLAG RELAYS						-RAR	RESISTORS									
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS									
	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING									
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE									
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION									
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD									
	C	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1							C	
		-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2								
-PFY		YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2									
-PGA		AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS									
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4									
	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5									
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES									
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS									
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS									
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS									
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS									
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES									
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS									
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS									
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES									
	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS									
	-PGV	VOLTMETERS						-TB	CONVERTER									
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE									
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE									
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE									
E	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING									
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE									
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER									
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)														F		
	-QBD4	TWO-WAY DISCONNECTOR																
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER																
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS		Scale	NTS		= H02		
					Prepared	YR	MV Switchgear for MNA Serang Plant				MVD-32 20kV Tank Farm WICA		Customer	PT Multimas Nabati Asahan		& 306		
	00	Issued for Approval	17.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-306.11		Sh. 5		
	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBC-W OUTGOING		Customer Doc. Nbr.			Cont. 27		
1		2		3		4		5		6		7		8				

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

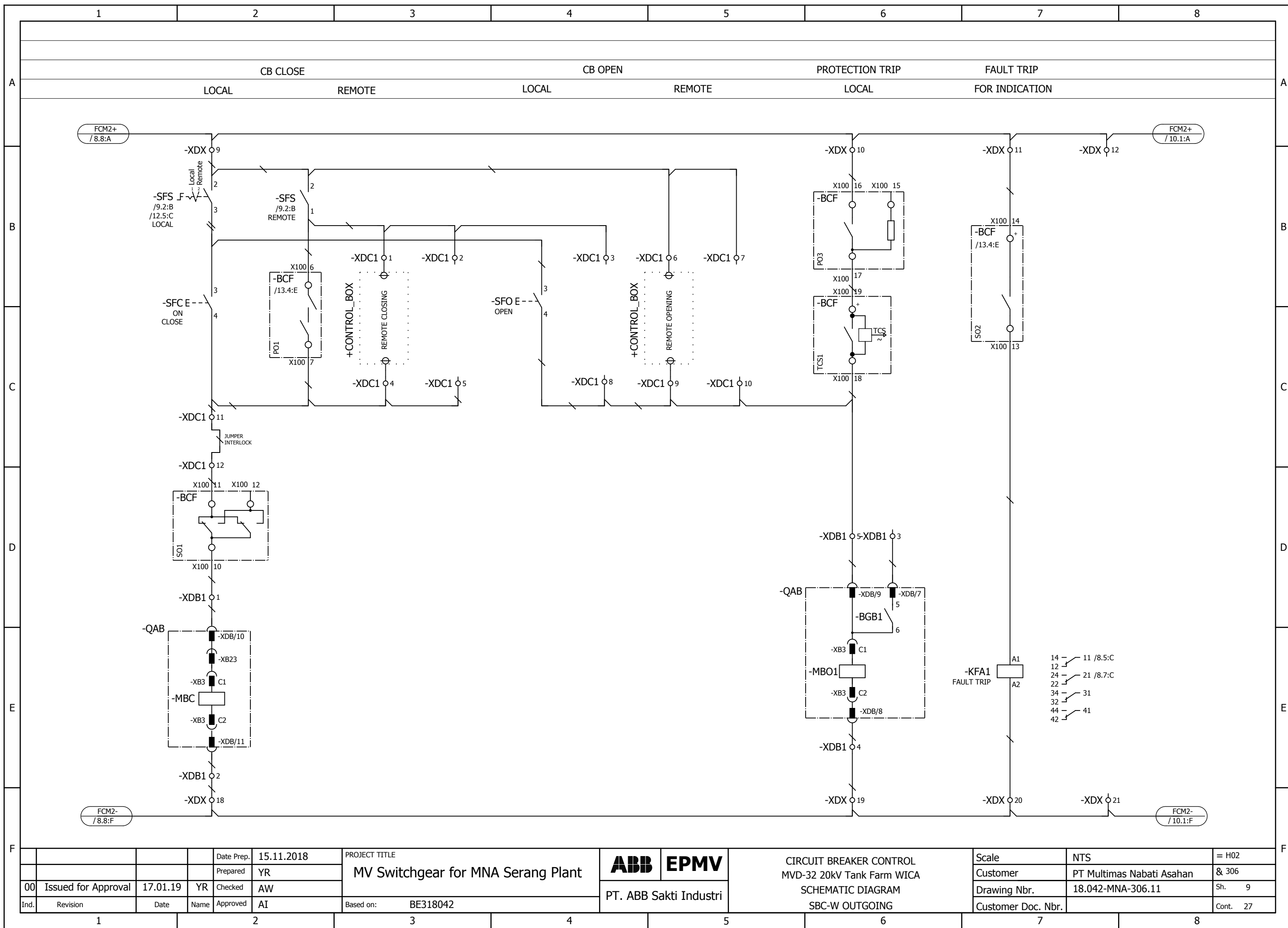
A	-TFA				ACTIVE POWER TRANSDUCERS				-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS				A		
	-TFC				CURRENT TRANSDUCERS					-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING					
	-TFF				FREQUENCY TRANSDUCERS				-XDI3		TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION					
	-TFJ				ACTIVE ENERGY TRANSDUCERS					-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS					
	-TFK				REACTIVE ENERGY TRANSDUCERS				-XDI6		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS					
	-TFM				MULTIFUNCTION TRANSDUCERS					-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS					
	-TFP				POWER-FACTOR TRANSDUCERS				-XDI8		TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS					
	-TFQ				REACTIVE POWER TRANSDUCERS					-XDI9	TERMINAL BLOCK FOR SPECIAL USE					
	-TFS				SIGNAL CONVERTERS				-XDJ		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD					
	-TFV				VOLTAGE TRANSDUCERS					-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS					
B	-WA				BUSBARS				-XDM		SEALABLE TERMINAL BLOCK FOR MEASUREMENT				B	
	-WBC				POWER CABLES					-XDS	CURRENT SOCKETS					
	-WE				EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.				-XDT		TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB					
	-WGC				CONTROL CABLES					-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK					
	-WGS				COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)				-XDU1		TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK					
	-WH				OPTICAL FIBRES					-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS					
	-XDA				TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS				-XDV1		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1					
	-XDA1				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1					-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2					
	-XDA2				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2				-XDV3		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3					
	-XDA3				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3					-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR					
	-XDA5				CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER				-XDX		SUPPORT TERMINAL BLOCK					
	-XDB				CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS					-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE					
	-XDB1				CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)				-XE2		EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE					
	-XDB2				CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER					-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE					
	-XDB10...-XDB89				CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS				-XE6		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE					
	-XDB7				CONNECTOR FOR TRUCK POSITION CONTACTS					-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE					
	-XDB91				CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR				-XE8		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE					
	-XDB92				CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR					-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT					
	-XDB93				CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR											
	-XDB95				CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET											
D	-XDC				CUSTOMER TERMINAL BLOCK											
	-XDC1				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER											
	-XDC2				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR											
	-XDC3				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT											
	-XDC4				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS											
	-XDC5				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER											
	-XDE				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE											
	-XDE1				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE											
	-XDE2				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2											
	-XDE3				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2											
	-XDF				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS											
	-XDF1				CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR											
	-XDF2				CONNECTOR FOR REAR VENTILATOR CIRCUITS											
	-XDF3				CONNECTOR FOR FRONT VENTILATOR CIRCUITS											
	-XDH				TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)											
	-XDI				TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)											
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB		EPMV	REFERENCE DESIGNATIONS MVD-32 20kV Tank Farm WICA		Scale	NTS	
					Prepared	YR				Customer				PT Multimas Nabati Asahan		& 306
	00	Issued for Approval	17.01.19	YR	Checked	AW				Drawing Nbr.				18.042-MNA-306.11		Sh. 6
	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBC-W OUTGOING		Customer Doc. Nbr.		Cont. 27
1		2		3		4		5		6		7		8		



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

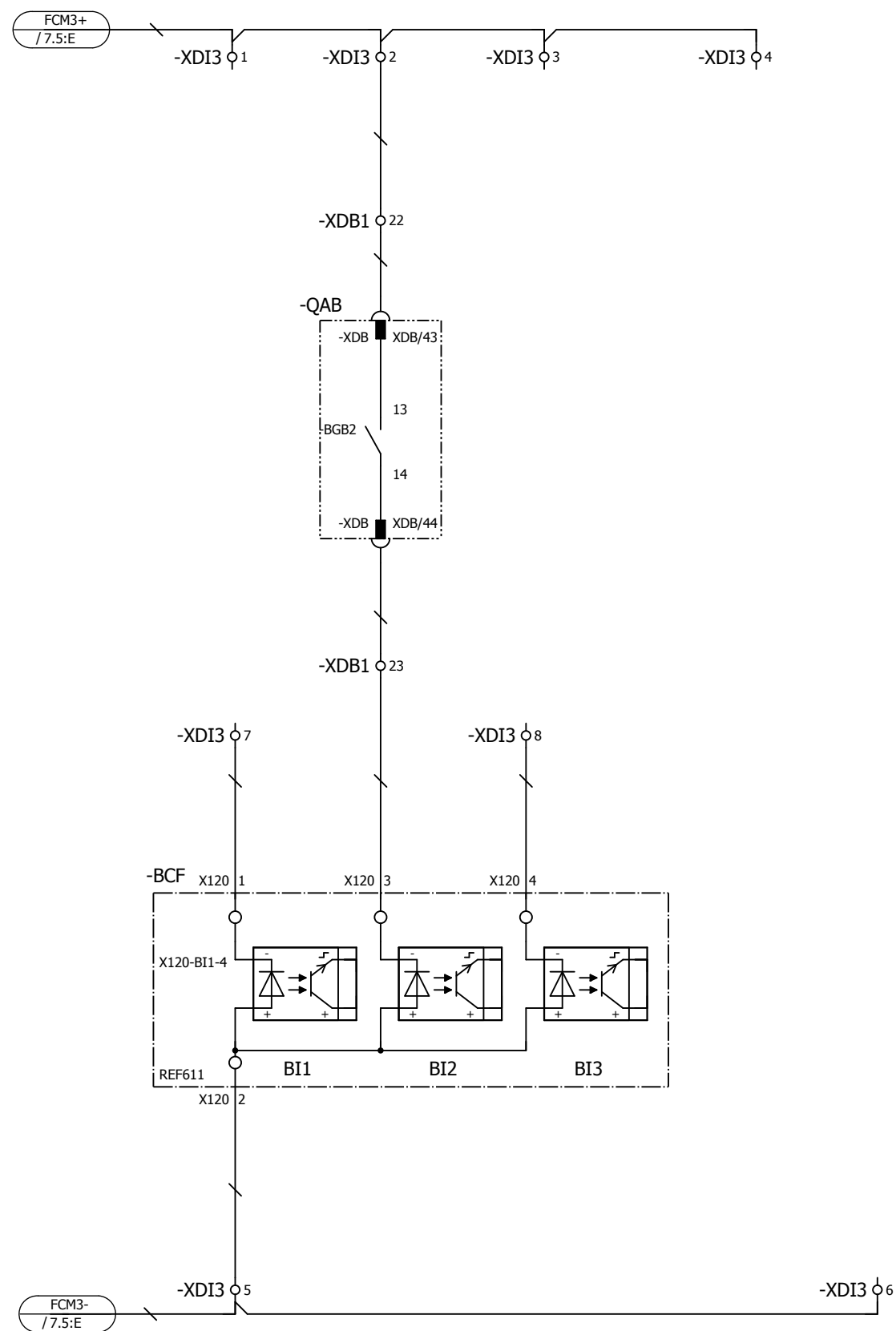


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



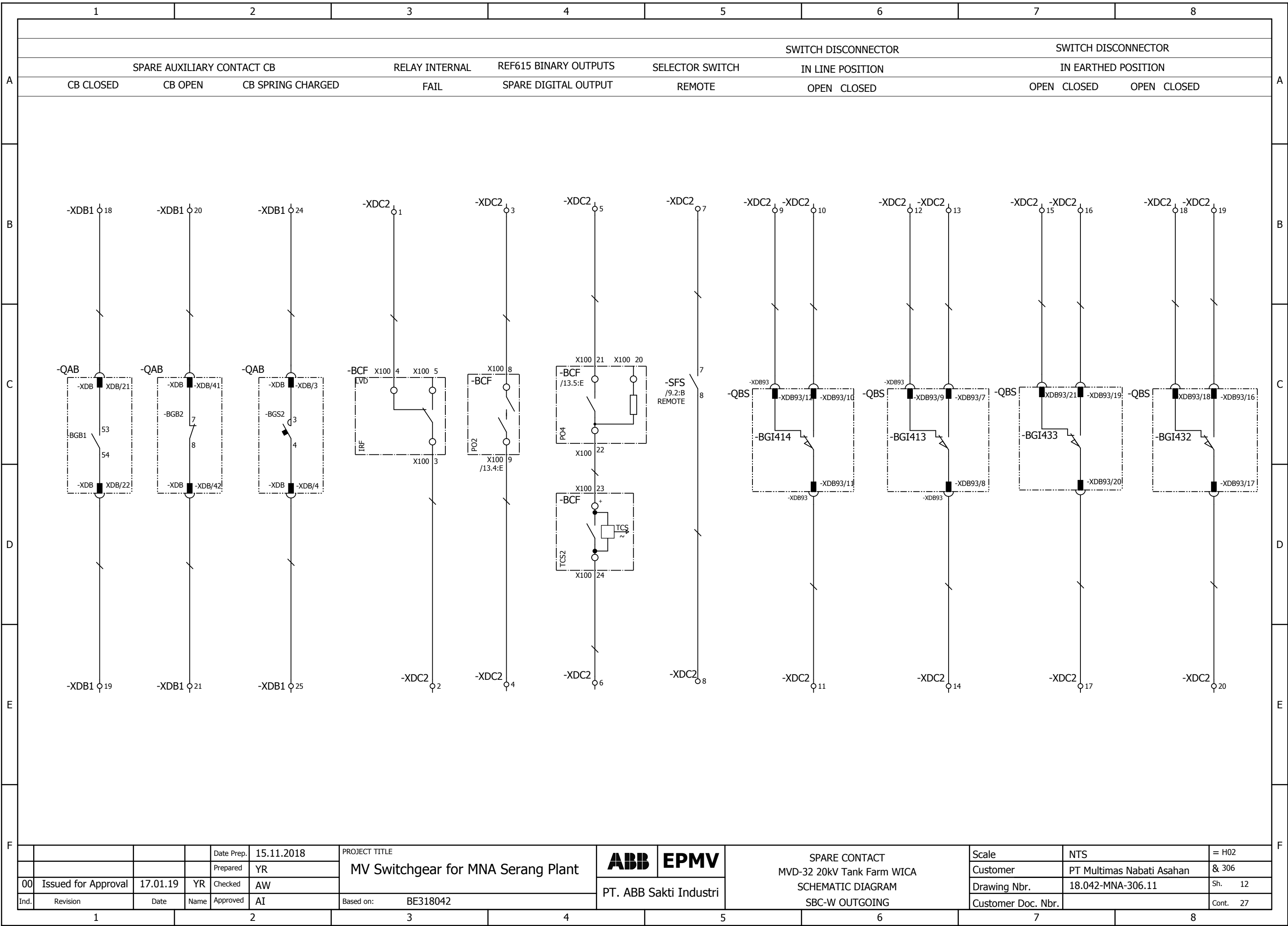
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY MOTOR & RELAY MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	17.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-306.11		Sh. 10		
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.			Cont. 27	
							PT. ABB Sakti Industri					

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

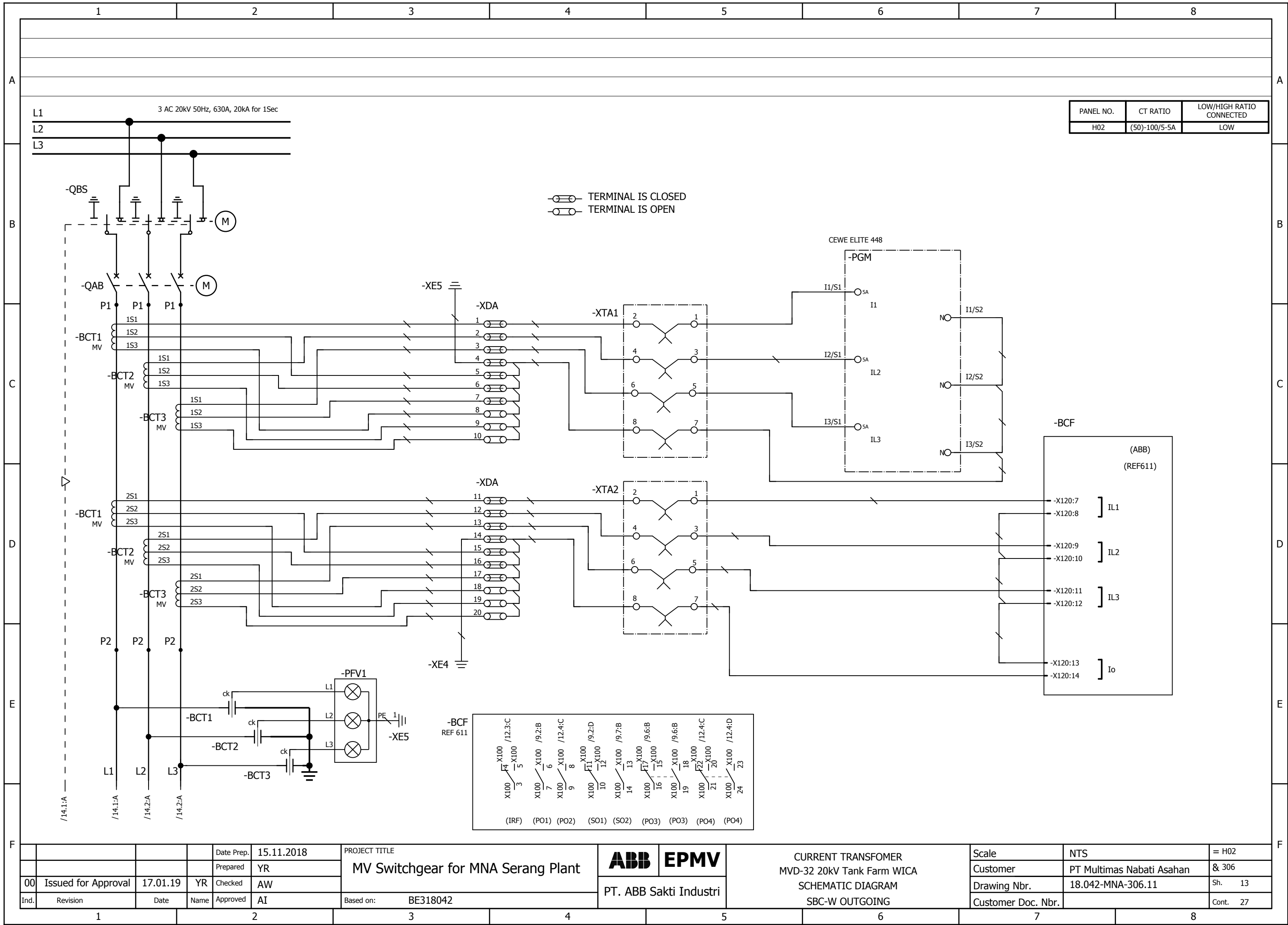


				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB EPMV	DIGITAL INPUT/OUTPUT RELAY MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02
				Prepared	YR				Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	17.01.19	YR	Checked	AW				Drawing Nbr.	18.042-MNA-306.11	Sh. 11
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 27

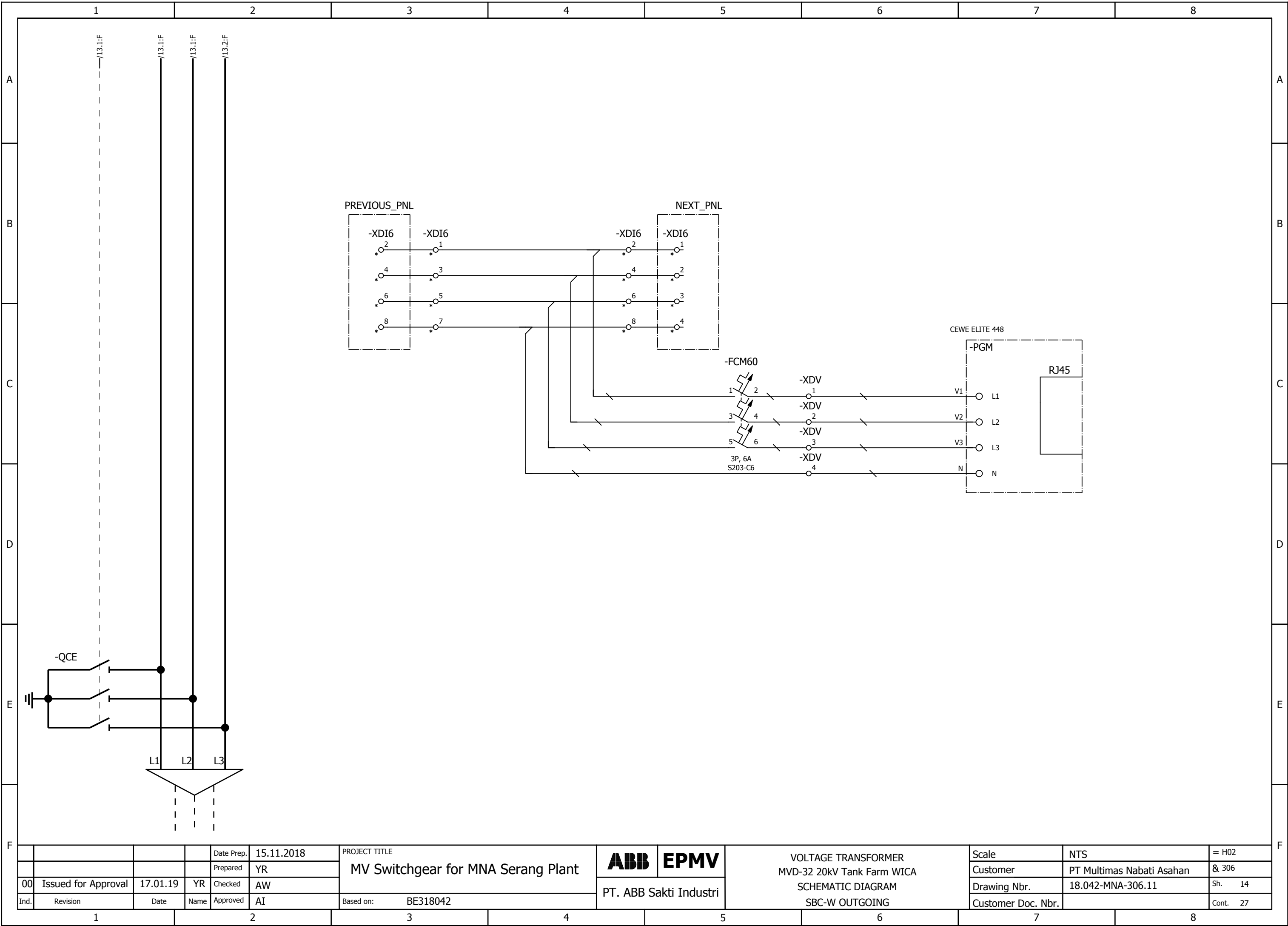
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



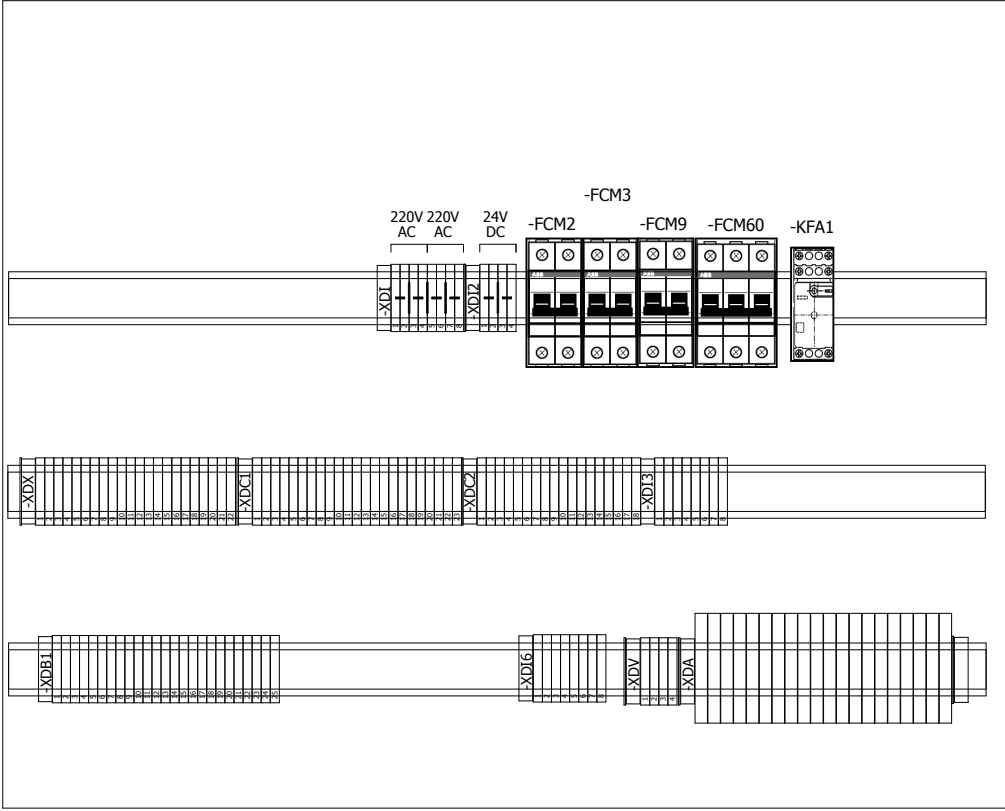
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-32 20kV Tank Farm WICA SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02
				Prepared	YR							Customer
00	Issued for Approval	17.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-306.11	Sh. 16
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 27

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

	1	2	3	4	5	6	7	8							
A	Summarized parts list							IDABB_F02_005-IDABB							
	Carry over: 0.00														
	TAG ID	DESCRIPTION			MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS							
	-BCF	Feeder Management Relay REF611HBAABC1NN11G			ABB	REF611HBAABC1NN11G	1	/8.2:C							
	-EB	HEATER 230VAC, 50W			SANYANG	CDXN010033P0009	1	/7.6:E							
	-FCM2	MCB S202-C10 MCB. 2P, 10A			ABB	2CDS252001R0104	1	/7.2:C							
	-FCM3;-FCM9	MCB S202-C6 MCB. 2P, 6A			ABB	2CDS252001R0064	2	/7.5:C;/7.6:C							
	-FCM60	MCB S203-C6 MCB. 3P, 6A			ABB	2CDS253001R0064	1	/14.5:C							
	-KFA1	Auxiliary Relay 230AC 4COS			PHOENIX	1YRA3DL266230001	1	/9.7:E							
	-PFG1	PILOT LIGHT GREEN #CL-523G#230VAC CL-523G			ABB	1SFA619402R5232	1	/8.5:E							
B	-PFR1	PILOT LIGHT RED #CL-523R#230VAC CL-523R			ABB	1SFA619402R5231	1	/8.4:E							
	-PFY1	PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y			ABB	1SFA619402R5233	1	/8.6:E							
	-PGM1	POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC			CEWE	1YRA4EA0350001	1	/8.3:C							
	-SFC	PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR			IDEC	1YRA3DA14700001	1	/9.2:B							
	-SFO	PUSH BUTTON, YW1B-M1E10-R RED COLOUR			IDEC	1YRA3DA14700002	1	/9.4:B							
C	-SFS	LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)			K&N	1YRA3DA17940011	1	/9.2:B							
	-XTA1;-XTA2	Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting			YONGSUNG	YSCTT-04C	2	/13.4:C;/13.4:D							
D															
E															
F															
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	PART LIST		Scale	NTS	= H02	
				Prepared	YR	MV Switchgear for MNA Serang Plant				MVD-32 20kV Tank Farm WICA		Customer	PT Multimas Nabati Asahan	& 306	
00	Issued for Approval	17.01.19	YR	Checked	AW			PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-306.11	Sh.	17	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			SBC-W OUTGOING		Customer Doc. Nbr.		Cont.	27	
	1	2		3		4		5		6		7		8	

F					E		D		C		B		A	
					REMARK	CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK
BLACK 2,5 mm ²					-XTA1:2				1	UT 6-T-HV P/P		-BCT1:1S1		BLACK 2,5 mm ²
BLACK 2,5 mm ²					-XTA1:4				2	UT 6-T-HV P/P		-BCT2:1S1		BLACK 2,5 mm ²
BLACK 2,5 mm ²					-XTA1:6				3	UT 6-T-HV P/P		-BCT3:1S1		BLACK 2,5 mm ²
BLACK 2,5 mm ²					-XTA1:8				● 4	UT 6-T-HV P/P		-XE5		Y/G 2,5 mm ²
BLACK 2,5 mm ²									● 5	UT 6-T-HV P/P		-BCT1:1S2		BLACK 2,5 mm ²
									● 6	UT 6-T-HV P/P		-BCT2:1S2		BLACK 2,5 mm ²
									● 7	UT 6-T-HV P/P		-BCT3:1S2		BLACK 2,5 mm ²
									● 8	UT 6-T-HV P/P		-BCT1:1S3		BLACK 2,5 mm ²
									● 9	UT 6-T-HV P/P		-BCT2:1S3		BLACK 2,5 mm ²
BLACK 2,5 mm ²									● 10	UT 6-T-HV P/P		-BCT3:1S3		BLACK 2,5 mm ²
BLACK 2,5 mm ²					-XTA2:2				11	UT 6-T-HV P/P		-BCT1:2S1		BLACK 2,5 mm ²
BLACK 2,5 mm ²					-XTA2:4				12	UT 6-T-HV P/P		-BCT2:2S1		BLACK 2,5 mm ²
BLACK 2,5 mm ²					-XTA2:6				13	UT 6-T-HV P/P		-BCT3:2S1		BLACK 2,5 mm ²
BLACK 2,5 mm ²					-XTA2:8				● 14	UT 6-T-HV P/P		-XE4		Y/G 2,5 mm ²
									● 15	UT 6-T-HV P/P		-BCT1:2S2		BLACK 2,5 mm ²
									● 16	UT 6-T-HV P/P		-BCT2:2S2		BLACK 2,5 mm ²
									● 17	UT 6-T-HV P/P		-BCT3:2S2		BLACK 2,5 mm ²
									● 18	UT 6-T-HV P/P		-BCT1:2S3		BLACK 2,5 mm ²
									● 19	UT 6-T-HV P/P		-BCT2:2S3		BLACK 2,5 mm ²
									● 20	UT 6-T-HV P/P		-BCT3:2S3		BLACK 2,5 mm ²

F										E										D										C										B										A																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							

F				E		D		C		B		A	

F										E		D		C		B		A	

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

F					E		D		C		B		A	